



Client Solution Design Document	
Client Name	Northwind Components
Project Name	NetSuite Support - 4521
Customization Name	Vendor Return Authorization Workflow

Basic Business Overview	Problem: Warehouse staff create vendor returns by hand, and approvals are tracked in email. Solution: A return authorization form with automated approval routing and a nightly status sync. Business Impact: Fewer lost returns and faster vendor credits.
Assumptions	Northwind will provide the approver list before development starts. Existing vendor records contain a valid email address.



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1 Business Context

1.1 Current State

- Warehouse staff email a spreadsheet to purchasing.
- Purchasing keys the return into a vendor bill credit.
- Approval status lives in email threads.

1.2 Desired State

- Warehouse staff submit returns from a single form.
- Approvers receive a task and respond in NetSuite.
- Return status is visible on the vendor record.

1.3 Open Questions

#	Question	Owner	Status
1	Which return reasons require a second approver?	Owner TBD	Open
2	Should returns above a dollar threshold route to finance?	Pat Lindqvist (Northwind)	Open

2. Scope

2.1 In Scope

- Return authorization form
- Approval routing
- Nightly status sync

2.2 Out of Scope

Not applicable to this solution.



3. Functional Requirements

FR-01 Return Intake

ID	Requirement
FR-01.1	The form requires a vendor, an item, a quantity, and a return reason.
FR-01.2	Quantity cannot exceed the quantity received on the original purchase order.
FR-01.3	Submitting the form creates a return record in Pending Approval status.

FR-02 Approval Routing

ID	Requirement
FR-02.1	A task is created for each approver on the return record.
FR-02.2	The return moves to Approved only when all approvers accept.
FR-02.3	A rejection returns the record to the requester with the rejection comment.

FR-03 Status Sync

ID	Requirement
FR-03.1	A nightly job updates the vendor record with the count of open returns.
FR-03.2	Closed returns are excluded from the open count.



4. Solution Design

4.1 Process Flow

The following depicts a complete process flow.



Edit in Lucid: <https://lucid.app/lucidchart/example-northwind-return>

4.2 Technical Approach

Return Authorization with Scripted Routing

A custom record holds the return. A user event script validates quantity and creates approval tasks. A scheduled script syncs status nightly.



Component Table

Component	Type	Purpose
Return Authorization	Custom Record	Holds the return header and lines.
Return Intake Form	Suitelet	Collects return details from the warehouse.
Return Validation	User Event	Validates quantity and creates tasks.

Key Logic Notes

- Quantity is compared with the purchase order received quantity at submit.
- Approval tasks are created once per approver.

4.3 Integrations and Dependencies

System	Relationship	Notes
Vendor email	Outbound notification	Sent from the NetSuite email template.

4.4 Script Outlines

EBS - Return Validation

Filename: ebs_ue_return_validation.js

Folder: SuiteScripts/Northwind/UserEvents

Script Type: User Event

Script ID: customscript_ebs_ue_return_validation

Script Parameters:

Name	Type	Value
Approver Role	List/Record	TBD
Max Quantity Variance	Integer	0
Notify Requester	Checkbox	Yes

Script Outline:

- **beforeSubmit(context):** Validates the quantity on each line.
 - Load the purchase order received quantity.
 - Reject the save when the quantity is higher.
- **afterSubmit(context):** Creates approval tasks.
 - Read the approver list from the parameter.



- o Create one task per approver.

EBS - Return Status Sync

Filename: ebs_ss_return_status_sync.js

Folder: SuiteScripts/Northwind/Scheduled

Script Type: Scheduled

Script ID: customscript_ebs_ss_return_status_sync

Script Outline:

- **execute(context):** Counts open returns per vendor and writes the count.
 - o Search open returns grouped by vendor.
 - o Update the vendor count field.

5. Data Model

5.1 New Custom Records

Return Authorization (customrecord_ebs_return_auth): one record per return with a link to the vendor and purchase order.

5.2 Key Custom Fields

The following custom fields must be added to support this solution.

Return Authorization

Field Label	Field ID	Type	Purpose
Vendor	custrecord_ebs_ra_vendor	List/Record	Vendor receiving the return.
Reason	custrecord_ebs_ra_reason	List/Record	Why the item is returned.

Vendor

Field Label	Field ID	Type	Purpose
Open Returns	custentity_ebs_open_returns	Integer Number	Count of open returns.



6. Approval and Sign-Off

By signing below, the approving party confirms they have reviewed this Solution Design Document and authorize Ethos Business Solutions to proceed with development based on the design described herein, subject to resolution of the open questions in Section 6 before the corresponding components are built.

Name	Organization	Role	Date	Decision
Matthew Garriga	Ethos Business Solutions	Technical Lead		
Pat Lindqvist	Northwind Components	Purchasing Manager		

Conditions:

- Approver list is confirmed before build starts.

7. Document Control

Date	Author	Version	Change Reference
10/07/2026	Matthew Garriga	1.0	Initial Draft